

PAPER_1548 — Vacuum Impedance $Z_0 = Z_0 = A_5 \cdot D_{BSFG} + S0_5 + D_{BSFG} + \Phi - F_{TRZ} \cdot \Phi - F_{TRZ}^2 \cdot SSQ \approx 376.75 \Omega$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Electromagnetism **Date:** June 18, 2026 **Status:** CLOSED — 0.005% closure

Observation

PAPER_1209DD_S613 (Electromagnetism Unified Proof Set) lists **Vacuum Impedance Z_0** as a closure with the formula:

Vacuum Impedance $Z_0 = Z_0 = A_5 \cdot D_{BSFG} + S0_5 + D_{BSFG} + \Phi - F_{TRZ} \cdot \Phi - F_{TRZ}^2 \cdot SSQ \approx 376.75 \Omega$

Status tier: **0.005%**. Units: Ω .

UQFF Closed Identity

$Z_0 = A_5 \cdot D_{BSFG} + S0_5 + D_{BSFG} + \Phi - F_{TRZ} \cdot \Phi - F_{TRZ}^2 \cdot SSQ \approx 376.75 \Omega \quad 0.005\%$

The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Vacuum Impedance Z_0 is a fundamental electromagnetism constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to “replace” the experimental value.

Reference

- Source: PAPER_1209DD_S613
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "z0_vacuum_impedance"})`

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